

Module 5 Homework

In this homework we'll put what we learned about Spark in practice.

For this homework we will be using the Yellow 2024-10 data from the official website:

```
wget https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2024-10.parquet
```

Question 1: Install Spark and PySpark

- Install Spark
- Run PySpark
- Create a local spark session
- Execute `spark.version`.

What's the output?

3.3.2

```
[9]: print(spark.version)

3.3.2
```

Note

To install PySpark follow this [guide](#)

Question 2: Yellow October 2024

```
micropython-partition_4_yellow_tripdata_2024-10.parquet
(base) > cd partition_4
(base) > ls
_SUCCESS
part-00000-ad65e590-4c32-48ae-9312-37d2bc47e8ee-c000.snappy.parquet
part-00001-ad65e590-4c32-48ae-9312-37d2bc47e8ee-c000.snappy.parquet
part-00002-ad65e590-4c32-48ae-9312-37d2bc47e8ee-c000.snappy.parquet
part-00003-ad65e590-4c32-48ae-9312-37d2bc47e8ee-c000.snappy.parquet
(base) > ls -lh part-00000-ad65e590-4c32-48ae-9312-37d2bc47e8ee-c000.snappy.parquet
-rw-r--r-- 1 eliza eliza 25M Mar  3 21:35 part-00000-ad65e590-4c32-48ae-9312-37d2bc47e8ee-c000.snappy.parquet
(base) >
```

(不知道为啥原来的源文件大小为62MB)

Read the October 2024 Yellow into a Spark Dataframe.

Repartition the Dataframe to 4 partitions and save it to parquet.

What is the average size of the Parquet (ending with .parquet extension) Files that were created (in MB)? Select the answer which most closely matches.

- 6MB
- **25MB**
- 75MB

- 100MB

Question 3: Count records

```
[17]: from pyspark.sql.functions import col, to_date

# Convert timestamp to date and filter only for October 15, 2024
df_filtered = df.filter(to_date(col("tpep_pickup_datetime")) == "2024-10-15")

# Count trips on that day
trip_count = df_filtered.count()
print(f"Total trips on October 15: {trip_count}")
```

Total trips on October 15: 128741

How many taxi trips were there on the 15th of October?

Consider only trips that started on the 15th of October.

- 85,567
- 105,567
- **125,567**
- 145,567

Question 4: Longest trip

```
[20]: from pyspark.sql.functions import col, unix_timestamp

df_with_duration = df.withColumn(
    "trip_duration_hours",
    (unix_timestamp(col("tpep_dropoff_datetime")) - unix_timestamp(col("tpep_pickup_datetime")))
)
max_duration = df_with_duration.agg({"trip_duration_hours": "max"}).collect()[0][0]
print(f"Longest trip duration: {max_duration} hours")
```

Longest trip duration: 162.61777777777777 hours

What is the length of the longest trip in the dataset in hours?

- 122
- 142
- **162**
- 182

Question 5: User Interface

Spark's User Interface which shows the application's dashboard runs on which local port?

- 80
- 443
- **4040**
- 8080

Question 6: Least frequent pickup location zone

```
[27]: # 按 PULocationID 分组并计算每个区域的出现频率
location_counts = df.groupBy("PULocationID").count()

# 找到出现频率最少的 PULocationID
least_frequent_location = location_counts.orderBy("count").first()

# 打印结果
print(f"The least frequent pickup location is Zone with PULocationID: {least_frequent_location}")
```

The least frequent pickup location is Zone with PULocationID: 105

```
[33]: # 假设 df_Lookup 包含 LocationID 和 Zone 的列

# 通过过滤查找特定的 LocationID 对应的 Zone
result = df_lookup.filter(df_lookup.LocationID == 105) \
                .select("Zone")

# 显示结果
result.show()
```

Zone
Governor's Island/...

```
[35]: # 查询并打印所有匹配的 Zone
result = df_lookup.filter(df_lookup.LocationID == 105).select("Zone").collect()
for row in result:
    print(row.Zone)
```

Governor's Island/Ellis Island/Liberty Island

Load the zone lookup data into a temp view in Spark:

```
wget https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv
```

Using the zone lookup data and the Yellow October 2024 data, what is the name of the LEAST frequent pickup location Zone?

- Governor's Island/Ellis Island/Liberty Island
- Arden Heights
- Rikers Island
- Jamaica Bay

Submitting the solutions

- Form for submitting: <https://courses.datatalks.club/de-zoomcamp-2025/homework/hw5>
- Deadline: See the website